Practice workbook · Area between a curve and the x-axis

Finding areas under one curve, splitting where it crosses the x-axis and adding the sizes of the separate pieces.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-19

1. Question 10 · part d · 4 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d only.

10

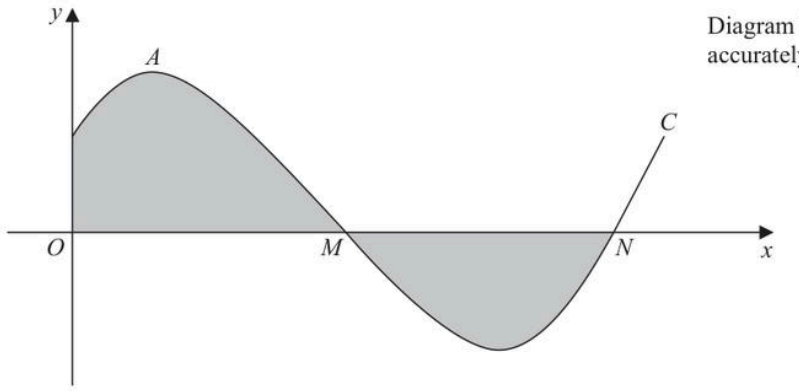


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows the curve C with equation $y = \frac{1}{2} + \sin 3x$ where $0 \leqslant x < \frac{2\pi}{3}$

The curve C crosses the x-axis at the points M and N

(a) Show that the coordinates of M are $\left(\frac{7\pi}{18}, 0\right)$ and find the coordinates of N

(3)

The curve C has a maximum at the point A

(b) Find the coordinates of A

(4)

(c) Find an equation of the tangent to C at M

Give your answer in the form $ay + b\sqrt{3}x - c\sqrt{3}\pi = 0$ where a, b and c are integers to be found.

(4)

The finite region, shown shaded in Figure 4, is bounded by the curve C, the y-axis and the part of the x-axis from O to N

(d) Use algebraic integration to find, to 3 significant figures, the total area of the shaded region.

(4)

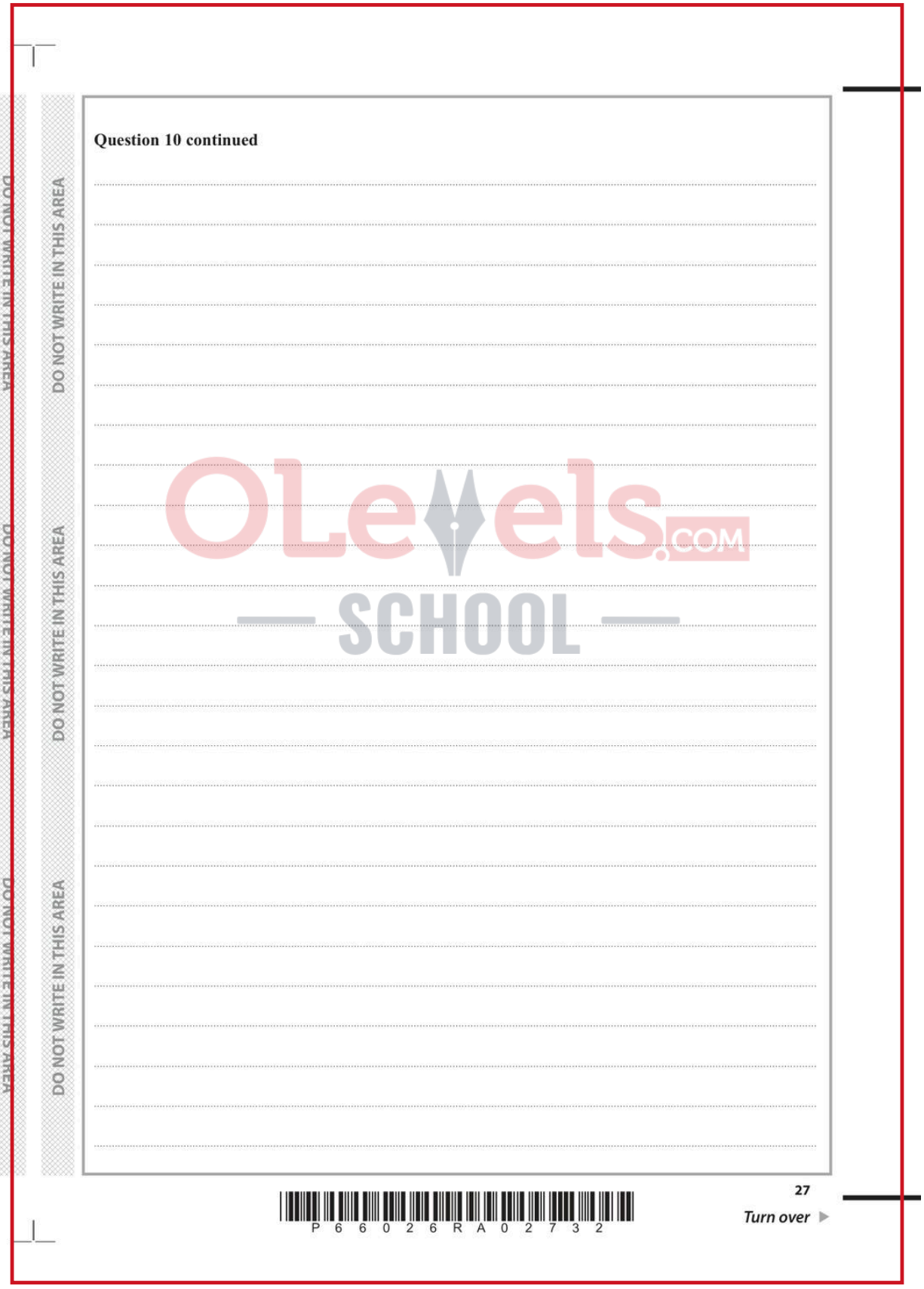
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1. Question 10 · part d · 4 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d only.



Question 10 continued

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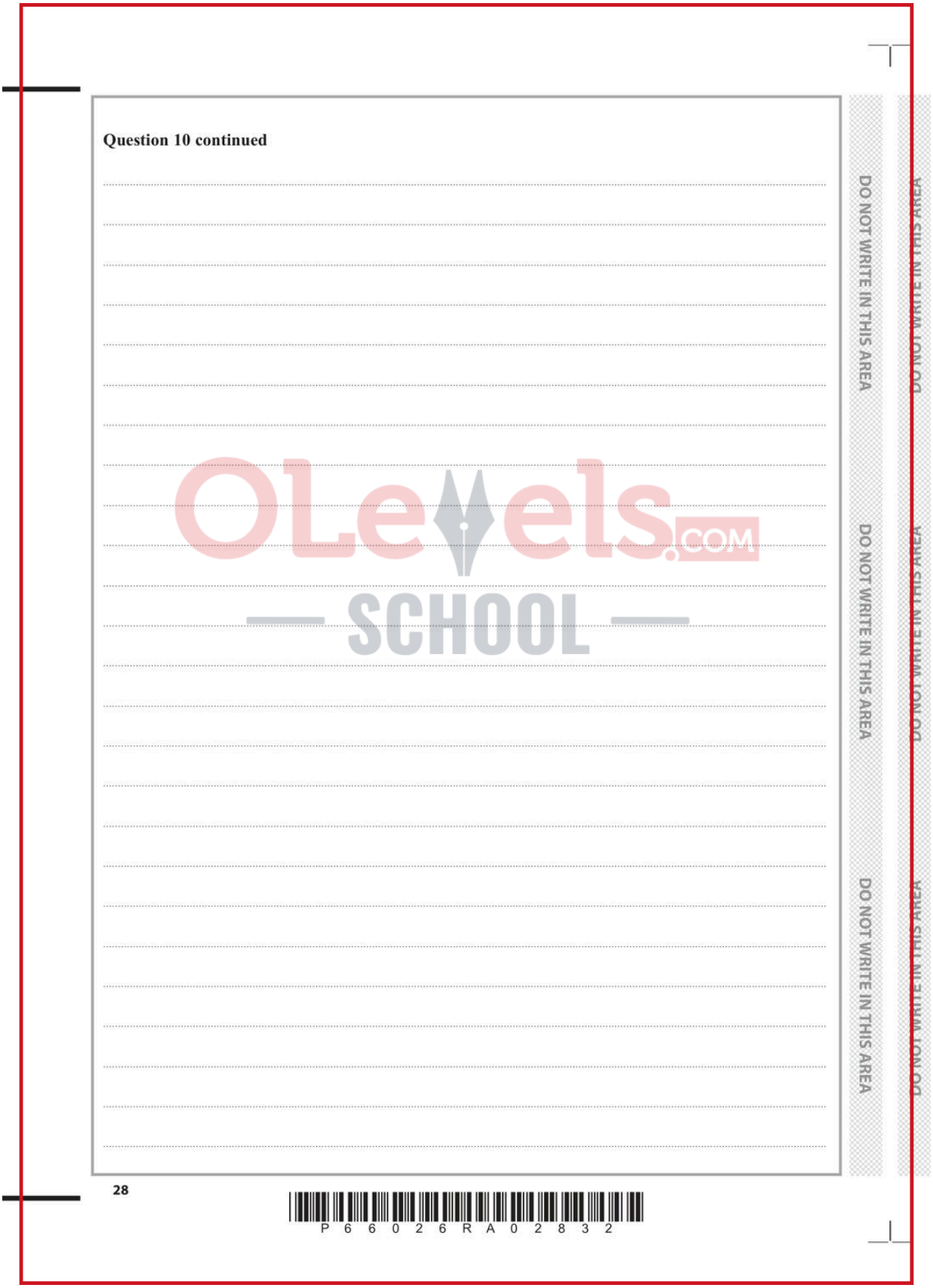
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Turn over ▶

1. Question 10 · part d · 4 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d only.



Question 10 continued

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1. Question 10 · part d · 4 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d only.

Question 10 continued

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(Total for Question 10 is 15 marks)

1. Question 10 · part d · 4 marks · Working and answer

Working and answer

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2. Question 9 · part c · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

9

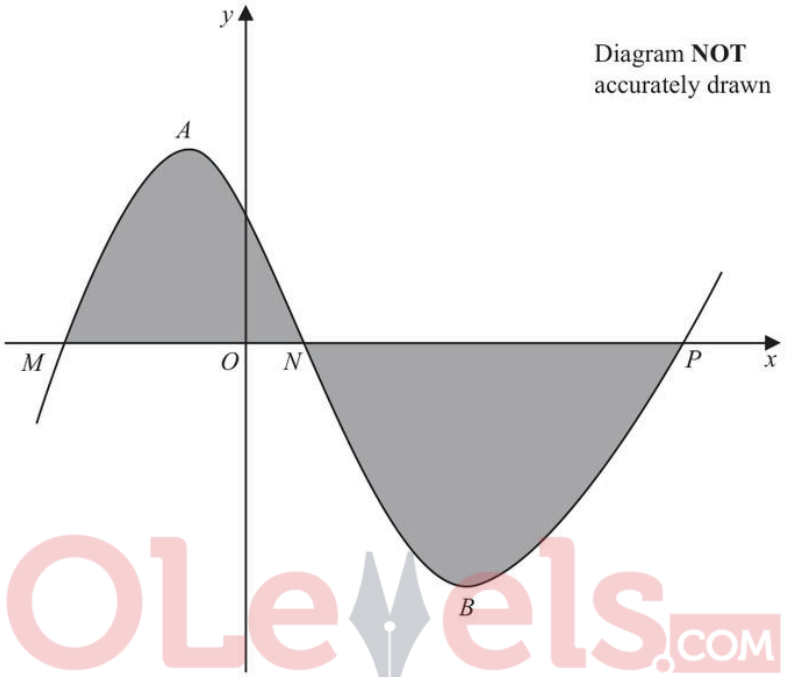


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$ where

$$f(x) = 2x^3 + ax^2 + bx + c$$

The curve C has a maximum at the point A with coordinates $\left(-\frac{1}{3}, \frac{100}{27}\right)$ and a minimum at the point B with coordinates $(2, -9)$

Given that a, b and c are integers

(a) show that $a = -5, b = -4$ and $c = 3$

(5)

(b) (i) Show that $(x+1)$ is a factor of $f(x)$

(1)

(ii) Hence, or otherwise, use algebra to factorise $f(x)$ completely.

(3)

The curve C crosses the x-axis at the points M, N and P
The finite regions shown shaded in Figure 4 are bounded by the curve C and parts of the x-axis from M to N and from N to P

(c) Use algebraic integration to determine the total area of the shaded regions. Give your answer as an exact fraction.

(4)

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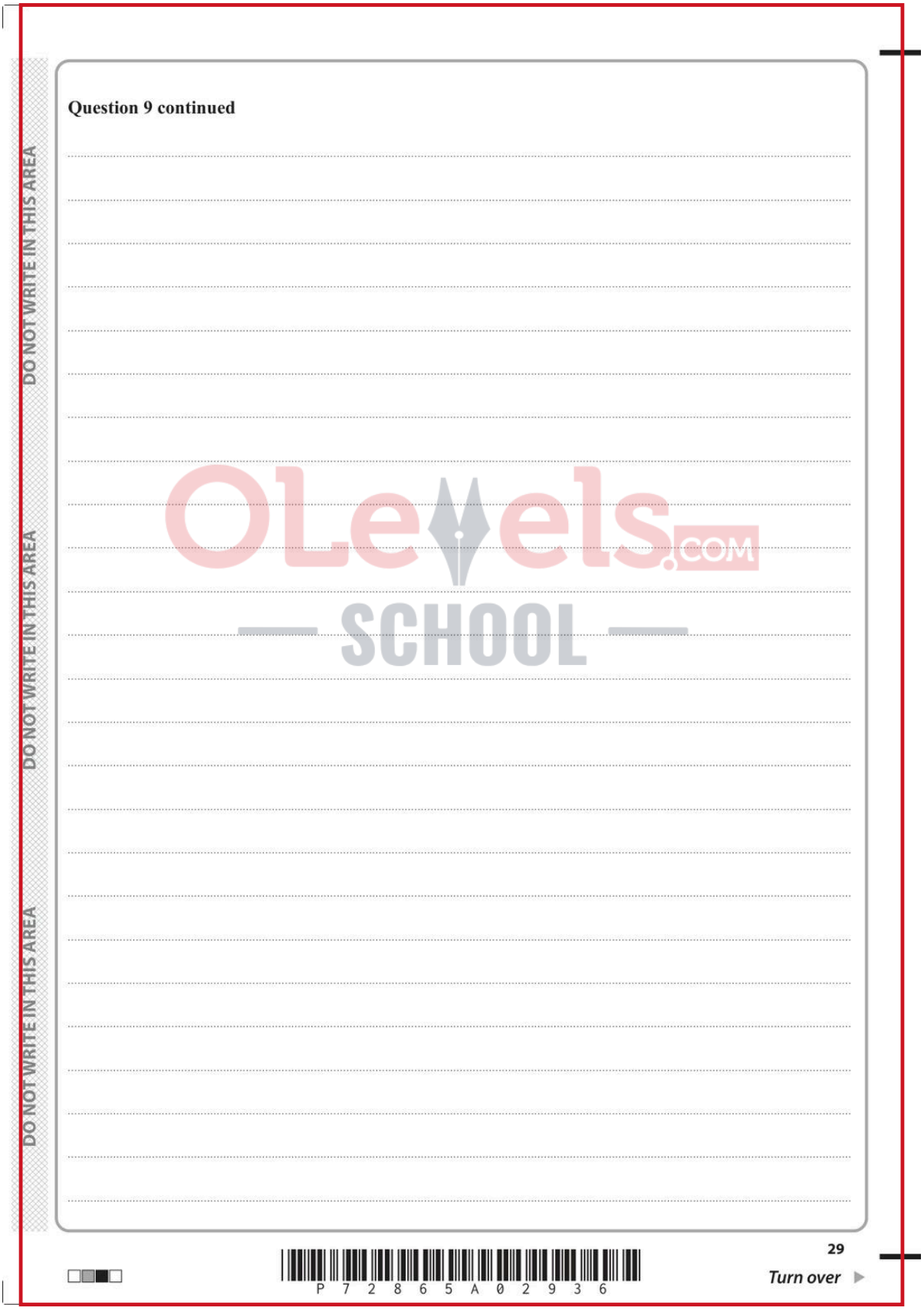
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P 7 2 8 6 5 A 0 2 8 3 6

2. Question 9 · part c · 4 marks

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Further practice · Attempt part c only.



2. Question 9 · part c · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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2. Question 9 · part c · 4 marks

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Further practice · Attempt part c only.

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Question 9 continued

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(Total for Question 9 is 13 marks)

2. Question 9 · part c · 4 marks · Working and answer

Working and answer

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3. Question 11 · part c · 3 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

11 $f(\theta) = (2\cos\theta - \sin\theta)(2\sin\theta + \cos\theta)$

(a) Show that $f(\theta) = \frac{3}{2}\sin 2\theta + 2\cos 2\theta$

(3)

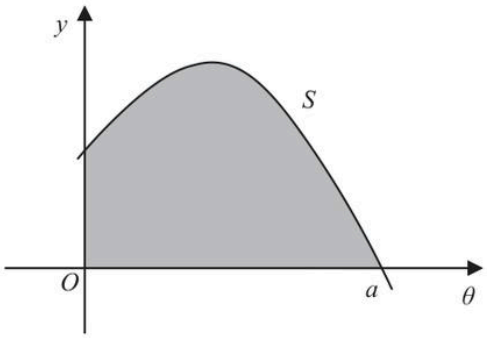


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows part of the curve S with equation $y = f(\theta) + 2$

Given that S intersects with the θ-axis at the point with coordinates $(a, 0)$

(b) using $\sin^2\theta + \cos^2\theta = 1$, or otherwise, show that $a = \frac{\pi}{2}$

(5)

(c) Using algebraic integration, find the exact area bounded by S, the positive θ-axis and the positive y-axis shown shaded in Figure 4

(3)

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3. Question 11 · part c · 3 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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3. Question 11 · part c · 3 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 11 marks)

3. Question 11 · part c · 3 marks · Working and answer

Working and answer

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4. Question 3 · part whole · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part whole only.

3 The curve C has equation $y = 9 - x^2$

Use algebraic integration to find the area of the finite region bounded by C and the x-axis.

(5)

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4. Question 3 · part whole · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 5 marks)

4. Question 3 · part whole · 5 marks · Working and answer

Working and answer

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5. Question 10 · part b · 8 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

10 Using suitable results for $\sin(A + B)$ and $\sin(A - B)$ from the formulae page,

(a) show that $2\sin 4x \cos x = \sin 5x + \sin 3x$

(3)

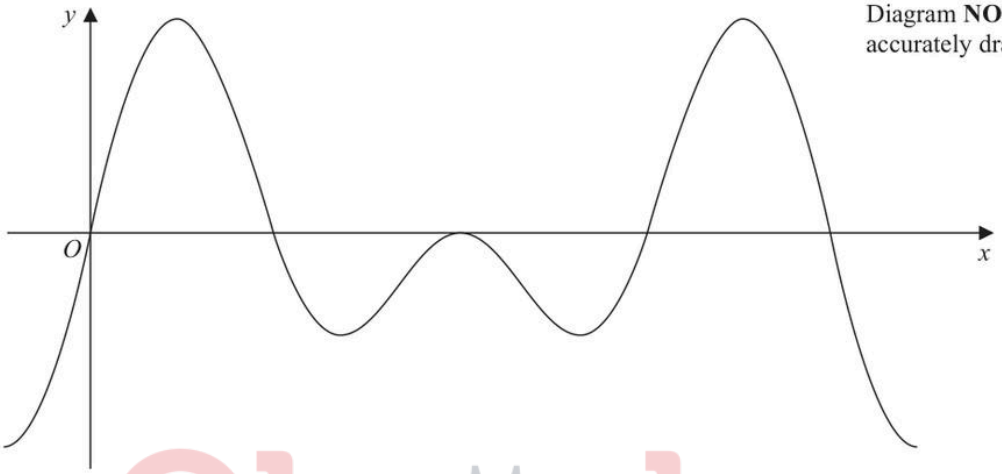


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows part of a sketch of the curve $y = 6\sin 4x \cos x$

(b) Using calculus, find the total area bounded by the curve and the x-axis between $x = 0$ and $x = \frac{\pi}{2}$

Give your answer to 3 significant figures.

(8)

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5. Question 10 · part b · 8 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 10 continued

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5. Question 10 · part b · 8 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 10 continued

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5. Question 10 · part b · 8 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 10 continued

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(Total for Question 10 is 11 marks)

5. Question 10 · part b · 8 marks · Working and answer

Working and answer

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6. Question 5 · part d · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

5 $f(x) = 2x^3 + ax^2 - 14x + b$ where a and b are constants.

When $f(x)$ is divided by $(x - 4)$ the remainder is 39

Given that $(x - 1)$ is a factor of $f(x)$

(a) show that $a = -3$ and find the value of b

(5)

(b) Hence factorise $f(x)$ completely.

(4)

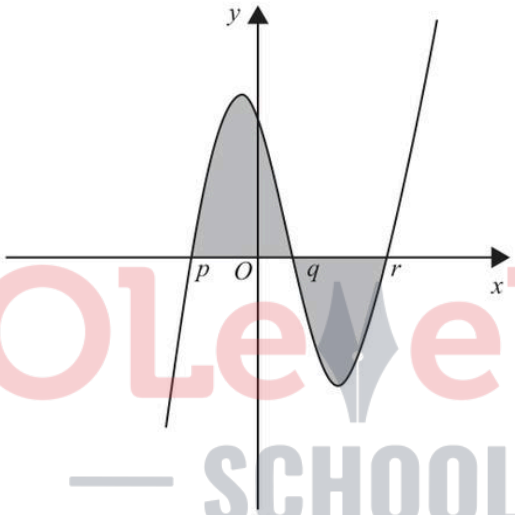


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows part of the curve C with equation $y = f(x)$

Given that C crosses the x-axis at the points with coordinates $(p, 0)$, $(q, 0)$ and $(r, 0)$

(c) write down the value of p, the value of q and the value of r

(3)

The region shown shaded in Figure 3 is bounded by the curve and the x-axis.

(d) Use algebraic integration to find the exact area of the shaded region.

(4)

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6. Question 5 · part d · 4 marks

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Further practice · Attempt part d only.

Question 5 continued

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6. Question 5 · part d · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 5 continued

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6. Question 5 · part d · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

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Question 5 continued

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(Total for Question 5 is 16 marks)

6. Question 5 · part d · 4 marks · Working and answer

Working and answer

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7. Question 7 · part c · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

7 $f(x) = 2x^3 + Ax^2 + Bx - 20$ where A and B are constants.

Given that

- $(2x+1)$ is a factor of $f(x)$
- when $f'(x)$ is divided by $(x-1)$ the remainder is -27

(a) show that $A = 3$ and find the value of B

(5)

(b) Hence, using algebra, solve the equation $f(x) = 0$

(3)

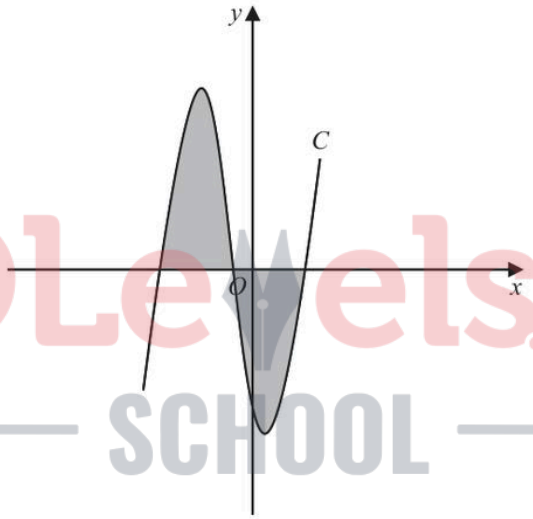


Figure 2

Figure 2 shows a sketch of part of the curve C with equation $y = f(x)$

(c) Use calculus to find the exact area of the finite region bounded by C and the x-axis, shown shaded in Figure 2

(4)

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7. Question 7 · part c · 4 marks

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Further practice · Attempt part c only.

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Question 7 continued

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7. Question 7 · part c · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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7. Question 7 · part c · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

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Question 7 continued

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(Total for Question 7 is 12 marks)

7. Question 7 · part c · 4 marks · Working and answer

Working and answer

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